Functional genomics reveals that Clostridium difficile Spo0A coordinates sporulation, virulence and metabolism.

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Clostridium difficile is an anaerobic, Gram-positive bacterium that can reside as a commensal within the intestinal microbiota of healthy individuals or cause life-threatening antibiotic-associated diarrhea in immunocompromised hosts. C. difficile can also form highly resistant spores that are excreted facilitating host-to-host transmission. The C. difficile spo0A gene encodes a highly conserved transcriptional regulator of sporulation that is required for relapsing disease and transmission in mice. Here we describe a genome-wide approach using a combined transcriptomic and proteomic analysis to identify Spo0A regulated genes. Our results validate Spo0A as a positive regulator of putative and novel sporulation genes as well as components of the mature spore proteome. We also show that Spo0A regulates a number of virulence-associated factors such as flagella and metabolic pathways including glucose fermentation leading to butyrate production. The C. difficile spo0A gene is a global transcriptional regulator that controls diverse sporulation, virulence and metabolic phenotypes coordinating pathogen adaptation to a wide range of host interactions. Additionally, the rich breadth of functional data allowed us to significantly update the annotation of the C. difficile 630 reference genome which will facilitate basic and applied research on this emerging pathogen.

Proteomic analysis of a NAP1 Clostridium difficile clinical isolate resistant to metronidazole.

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Clostridium difficile is an anaerobic, Gram-positive bacterium that has been implicated as the leading cause of antibiotic-associated diarrhea. Metronidazole is currently the first-line treatment for mild to moderate C. difficile infections. Our laboratory isolated a strain of C. difficile with a stable resistance phenotype to metronidazole. A shotgun proteomics approach was used to compare differences in the proteomes of metronidazole-resistant and -susceptible isolates. NAP1 C. difficile strains CD26A54_R (Met-resistant), CD26A54_S (reduced- susceptibility), and VLOO13 (Met-susceptible) were grown to mid-log phase, and spiked with metronidazole at concentrations 2 doubling dilutions below the MIC. Peptides from each sample were labeled with iTRAQ and subjected to 2D-LC-MS/MS analysis. In the absence of metronidazole, higher expression was observed of some proteins in C. difficile strains CD26A54_S and CD26A54_R that may be involved with reduced susceptibility or resistance to metronidazole, including DNA repair proteins, putative nitroreductases, and the ferric uptake regulator (Fur). After treatment with metronidazole, moderate increases were seen in the expression of stress-related proteins in all strains. A moderate increase was also observed in the expression of the DNA repair protein RecA in CD26A54_R. This study provided an in-depth proteomic analysis of a stable, metronidazole-resistant C. difficile isolate. The results suggested that a multi-factorial response may be associated with high level metronidazole-resistance in C. difficile, including the possible roles of altered iron metabolism and/or DNA repair.

Para-cresol production by Clostridium difficile affects microbial diversity and membrane integrity of Gram-negative bacteria.

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Clostridium difficile is a Gram-positive spore-forming anaerobe and a major cause of antibiotic-associated diarrhoea. Disruption of the commensal microbiota, such as through treatment with broad-spectrum antibiotics, is a critical precursor for colonisation by C. difficile and subsequent disease. Furthermore, failure of the gut microbiota to recover colonisation resistance can result in recurrence of infection. An unusual characteristic of C. difficile among gut bacteria is its ability to produce the bacteriostatic compound para-cresol (p-cresol) through fermentation of tyrosine. Here, we demonstrate that the ability of C. difficile to produce p-cresol in vitro provides a competitive advantage over gut bacteria including Escherichia coli, Klebsiella oxytoca and Bacteroides thetaiotaomicron. Metabolic profiling of competitive co-cultures revealed that acetate, alanine, butyrate, isobutyrate, p-cresol and p-hydroxyphenylacetate were the main metabolites responsible for differentiating the parent strain C. difficile (630Δerm) from a defined mutant deficient in p-cresol production. Moreover, we show that the p-cresol mutant displays a fitness defect in a mouse relapse model of C. difficile infection (CDI). Analysis of the microbiome from this mouse model of CDI demonstrates that colonisation by the p-cresol mutant results in a distinctly altered intestinal microbiota, and metabolic profile, with a greater representation of Gammaproteobacteria, including the Pseudomonales and Enterobacteriales. We demonstrate that Gammaproteobacteria are susceptible to exogenous p-cresol in vitro and that there is a clear divide between bacterial Phyla and their susceptibility to p-cresol. In general, Gram-negative species were relatively sensitive to p-cresol, whereas Gram-positive species were more tolerant. This study demonstrates that production of p-cresol by C. difficile has an effect on the viability of intestinal bacteria as well as the major metabolites produced in vitro. These observations are upheld in a mouse model of CDI, in which p-cresol production affects the biodiversity of gut microbiota and faecal metabolite profiles, suggesting that p-cresol production contributes to C. difficile survival and pathogenesis.

Identification of Functional Spo0A Residues Critical for Sporulation in Clostridioides difficile.

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Clostridioides difficile is an anaerobic, Gram-positive pathogen that is responsible for C. difficile infection (CDI). To survive in the environment and spread to new hosts, C. difficile must form metabolically dormant spores. The formation of spores requires activation of the transcription factor Spo0A, which is the master regulator of sporulation in all endospore-forming bacteria. Though the sporulation initiation pathway has been delineated in the Bacilli, including the model spore-former Bacillus subtilis, the direct regulators of Spo0A in C. difficile remain undefined. C. difficile Spo0A shares highly conserved protein interaction regions with the B. subtilis sporulation proteins Spo0F and Spo0A, although many of the interacting factors present in B. subtilis are not encoded in C. difficile. To determine if comparable Spo0A residues are important for C. difficile sporulation initiation, site-directed mutagenesis was performed at conserved receiver domain residues and the effects on sporulation were examined. Mutation of residues important for homodimerization and interaction with positive and negative regulators of B. subtilis Spo0A and Spo0F impacted C. difficile Spo0A function. The data also demonstrated that mutation of many additional conserved residues altered C. difficile Spo0A activity, even when the corresponding Bacillus interacting proteins are not apparent in the C. difficile genome. Finally, the conserved aspartate residue at position 56 of C. difficile Spo0A was determined to be the phosphorylation site that is necessary for Spo0A activation. The finding that Spo0A interacting motifs maintain functionality suggests that C. difficile Spo0A interacts with yet unidentified proteins that regulate its activity and control spore

formation.

Clostridioides difficile: Characterization of the circulating toxinotypes in an Argentinean public hospital.

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Clostridioides difficile is a spore-forming anaerobe microorganism associated to nosocomial diarrhea. Its virulence is mainly associated with TcdA and TcdB toxins, encoded by their respective tcdA and tcdB genes. These genes are part of the pathogenicity locus (PaLoc). Our aim was to characterize relevant C. difficile toxinotypes circulating in the hospital setting. The tcdA and tcdB genes were amplified and digested with different restriction enzymes: EcoRI for tcdA; HincII and AccI for tcdB. In addition, the presence of the cdtB (binary toxin) gene, TcdA and TcdB toxins by dot blot and the cytotoxic effect of culture supernatants on Vero cells, were evaluated. Altogether, these studies revealed three different circulating toxinotypes according to Rupnik's classification: 0, I and VIII, being the latter the most prevalent one. Even though more studies are certainly necessary (e.g. sequencing analysis), it is worth noting that the occurrence of toxinotype I could be related to the introduction of bacteria from different geographical origins. The multivariate analysis conducted on the laboratory values of individuals infected with the most prevalent toxinotype (VIII) showed that the isolates associated with fatal outcomes (GCD13, GCD14 and GCD22) are located in regions of the biplots related to altered laboratory values at admission. In other patients, although laboratory values at admission were not correlated, levels of urea, creatinine and white blood cells were positively correlated after the infection was diagnosed. Our study reveals the circulation of different toxinotypes of C. difficile strains in this public hospital. The variety of toxinotypes can arise from pre-existing microorganisms as well as through the introduction of bacteria from other geographical regions. The existence of microorganisms with different pathogenic potential is relevant for the control, follow-up, and treatment of the infections.

Clostridioides difficile Infections: Prevention and Treatment Strategies.

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Clostridioides difficile is the most common causative agent of antibiotic-associated diarrhea. This spore forming, obligate anaerobic, gram-positive bacillus is becoming responsible for an increasing number of infections worldwide, both in community and in hospital settings, whose severity can vary widely from an asymptomatic infection to a lethal disease. While discontinuation of antimicrobial agents and antibiotic treatment of the infection remain the cornerstone of therapy, more recent fecal microbiota transplantation has also been valid as a therapy. The use of probiotics, especially Saccharomyces boulardii CNCM I-745 have become valid forms of prevention therapy. Although there are studies in adults with microbiota-targeted new generation therapies and Clostridium difficile vaccines, there are no data in the paediatric age group yet.

Influence of microbiota on the growth and gene expression of Clostridioides difficile in an in vitro coculture model.

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Clostridioides difficile is an anaerobic, spore-forming, Gram-positive pathogenic bacterium. This study aimed to analyze the effect of two samples of healthy fecal microbiota on C. difficile gene expression and growth using an in vitro coculture model. The inner compartment was cocultured with spores of the C. difficile polymerase chain reaction (PCR)-ribotype 078, while the outer compartment contained fecal samples from donors to mimic the microbiota (FD1 and FD2). A fecal-free plate served as a control (CT). RNA-Seq and quantitative PCR confirmation were performed on the inner compartment sample. Similarities in gene expression were observed in the presence of the microbiota. After 12 h, the expression of genes associated with germination, sporulation, toxin production, and growth was downregulated in the presence of the microbiota. At 24 h, in an iron-deficient environment, C. difficile activated several genes to counteract iron deficiency. The expression of genes associated with germination and sporulation was upregulated at 24 h compared with 12 h in the presence of microbiota from donor 1 (FD1). This study confirmed previous findings that C. difficile can use ethanolamine as a primary nutrient source. To further investigate this interaction, future studies will use a simplified coculture model with an artificial bacterial consortium instead of fecal samples.

Analysis of factors affecting the length of hospitalization of patients with Clostridioides difficile infection: a cross-sectional study.

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Clostridioides difficile infection (CDI) is an infectious disease caused by the gram-positive, anaerobic bacterium C. difficile. The vulnerable populations for CDI include the elderly, immunocompromised individuals, and hospitalized patients, especially those undergoing antimicrobial therapy, which is a significant risk factor for this infection. Due to its complications and increased resistance to treatment, CDI often leads to longer hospital stays. This study aimed to determine the average length of hospital stay (LOS) of Polish patients with CDI and to identify factors affecting the LOS of infected patients. The study analyzed medical records of adult patients treated with CDI in one of the biggest clinical hospitals in Poland between 2016-2018. Information encompassed the patient's age, LOS results of selected laboratory tests, number of antibiotics used, nutritional status based on Nutritional Risk Screening (NRS 2002), year of hospitalization, presence of diarrhea on admission, systemic infections, additional conditions, and undergone therapies. The systematic collection of these variables forms the foundation for a comprehensive analysis of factors influencing the length of stay. In the study period, 319 patients with CDI were hospitalized, with a median LOS of 24 days (min-max = 2-344 days). The average LOS was 4.74 days in 2016 (median = 28 days), 4.27 days in 2017 (median = 24 days), and 4.25 days in 2018 (median = 23 days). There was a weak negative correlation (Rho = -0.235, $p < 0.001$) between albumin level and LOS and a weak positive correlation between NRS and LOS (Rho = 0.219, $p < 0.001$). Patients admitted with diarrhea, a history of stroke or pneumonia, those taking certain antibiotics (penicillins, cephalosporins, carbapenems, fluoroquinolones, aminoglycosides, colistin), and those using proton pump inhibitors, exhibited longer hospitalizations (all p 0.05). However, the multivariate regression model explained a substantial portion of the variance in hospitalization length, with an R-squared value of 0.844. Hospitalization of a patient with CDI is long. Low albumin levels and increased risk of malnutrition were observed in longer hospitalized patients. Longer hospitalized patients had pneumonia, stroke, or surgery, and were admitted for a reason other than CDI.

Clostridioides difficile SinR' regulates toxin, sporulation and motility through protein-protein interaction with SinR.

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Clostridioides difficile is a Gram-positive, anaerobic bacterium. It is known that C. difficile is one of the major causes of antibiotic associated diarrhea. The enhanced antibiotic resistance observed in C. difficile is the result of highly resistant spores produced by the bacterium. In Bacillus subtilis, the sin operon is involved in sporulation inhibition. Two proteins coded within this operon, SinR and SinI, have an antagonistic relationship; SinR acts as an inhibitor to sporulation whereas SinI represses the activity of SinR, thus allowing the bacterium to sporulate. In a previous study, we examined the sin locus in C. difficile and named the two genes associated with this operon sinR and sinR', analogous to sinR and sinI in B. subtilis, respectively. We have shown that SinR and SinR' have pleiotropic roles in pathogenesis pathways and interact antagonistically with each other. Unlike B. subtilis SinI, SinR' in C. difficile carries two domains: the HTH domain and the Multimerization Domain (MD). In this study, we first performed a GST Pull-down experiment to determine the domain within SinR' that interacts with SinR. Second, the effect of these two domains on three phenotypes; sporulation, motility, and toxin production was examined. The findings of this study confirmed the prediction that the Multimerization Domain (MD) of SinR' is responsible for the interaction between SinR and SinR'. It was also discovered that SinR' regulates sporulation, toxin production and motility primarily by inhibiting SinR activity through the Multimerization Domain (MD).

Clostridioides difficile spores stimulate inflammatory cytokine responses and induce cytotoxicity in macrophages.

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Clostridioides difficile is a gram-positive, spore-forming anaerobic bacterium, and the leading cause of antibiotic-associated diarrhea worldwide. During C. difficile infection, spores germinate in the presence of bile acids into vegetative cells that subsequently colonize the large intestine and produce toxins. In this study, we demonstrated that C. difficile spores can universally adhere to, and be phagocytosed by, murine macrophages. Only spores from toxigenic strains were able to significantly stimulate the production of inflammatory cytokines by macrophages and subsequently induce significant cytotoxicity. Spores from the isogenic TcdA and TcdB double mutant induced significantly lower inflammatory cytokines and cytotoxicity in macrophages, and these activities were restored by pre-exposure of the spores to either toxins. These findings suggest that during sporulation, spores might be coated with C. difficile toxins from the environment, which could affect C. difficile pathogenesis in vivo.